INCH-POUND
MS51520C
w/ AMENDMENT 1
29 May 2020
SUPERSEDING
MS51520C
15 November 2017

DETAIL SPECIFICATION SHEET

NIPPLE, TUBE, BULKHEAD
37 DEGREE FLARED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Reactivated after 15 November 2017 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.

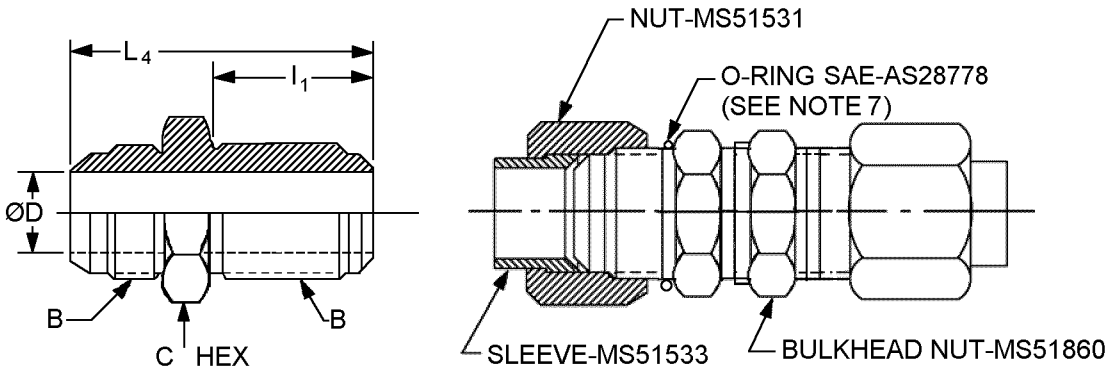


FIGURE 1. Nipple tube.

AMSC N/A

FSC 4730

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Suffix Designator (see note 8)		Tube OD nom.	B Straight thread	C Hex flat Nom.	D diameter	
Assembly	Body				Basic inches (mm)	Tolerance inches (mm)
A2	B2	1/8	.3125-24 UNF-2A	9/16	.062 (1.57)	±.004 (0.10)
A3	B3	3/16	.3750-24 UNF-2A	5/8	.125 (3.18)	
A4	B4	1/4	.4375-20 UNF-2A	11/16	.172 (4.37)	
A5	B5	5/16	.5000-20 UNF-2A	3/4	.234 (5.94)	
A6	B6	3/8	.5625-18 UNF-2A	13/16	.297 (7.54)	±.008 (0.20)
A8	B8	1/2	.7500-16 UNF-2A	1	.391 (9.93)	
A10	B10	5/8	.8750-14 UNF-2A	1 1/8	.484 (12.29)	
A12	B12	3/4	1.0625-12 UN-2A	1 3/8	.609 (15.47)	
A14	B14	7/8	1.1875-12 UN-2A	1 1/2	.718 (18.24)	
A16	B16	1	1.3125-12 UN-2A	1 5/8	.844 (21.44)	
A20	B20	1 1/4	1.6250-12 UN-2A	1 7/8	1.078 (27.38)	
A24	B24	1 1/2	1.8750-12 UN-2A	2 1/8	1.312 (33.32)	±.012 (0.31)
A32	B32	2	2.500-12 UN-2A	2 3/4	1.781 (45.24)	

Suffix designator		l_1 inches (mm) ±.020 (0.51)	L_4 inches (mm) ±.020 (0.51)
Assembly	Body		
A2	B2	1.110 (28.19)	1.870 (47.50)
A3	B3	1.110 (28.19)	1.900 (48.26)
A4	B4	1.200 (30.48)	2.070 (52.58)
A5	B5	1.200 (30.48)	2.070 (52.58)
A6	B6	1.280 (32.51)	2.180 (55.37)
A8	B8	1.440 (36.58)	2.440 (61.98)
A10	B10	1.580 (40.13)	2.740 (69.60)
A12	B12	1.750 (44.45)	3.090 (78.49)
A14	B14	1.750 (44.45)	3.120 (79.25)
A16	B16	1.750 (44.45)	3.140 (79.76)
A20	B20	1.800 (45.72)	3.310 (84.07)
A24	B24	1.810 (45.97)	3.520 (89.41)
A32	B32	2.090 (53.09)	4.200 (106.68)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges; remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514 for 37° flared fittings.
5. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
6. Threads shall be in accordance with ASME B1.1.
7. Procure the O-rings separately and install before installation of the fittings.
8. Assemblies are only furnished to this specification (body, nuts, and sleeves). Bodies are not to be stocked, stored, or issued.

FIGURE 1. Plug - Continued.

REQUIREMENTS:

Fittings shall be specified on figure 1 and in table I.

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Components of the assemblies shall be of the same material and finish.

Materials shall be in accordance with MIL-DTL-18866.

Finishes shall be in accordance with table I.

Unless otherwise specified, all plating's shall be capable of meeting 96-hour salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

TABLE I. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 2 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ 7/
H	Steel	Zinc-aluminum in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/
KA	Steel	Zinc Nickel Electroplate, MIL-PRF-32647, Type K20 torque/tension modifier.3/ 4/ 8/
KB	Steel	Zinc Nickel Electroplate, MIL-PRF-32647, Type K16 torque/tension modifier. 3/ 4/ 8/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance with MIL-DTL-16232 type Z, class 1.
R	Steel	Zinc plating in accordance with ASTM B633, type V, Fe/Zn 8. 4/
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 6/ 9/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 6/ 9/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8. 4/ 5/
ZC	Steel	Zinc may be any zinc plating from PIN codes J and R with a colored chromate coating. 4/ 5/
ZF	Steel	Zinc plating in accordance with ASTM B633, type VI, class 8. 4/

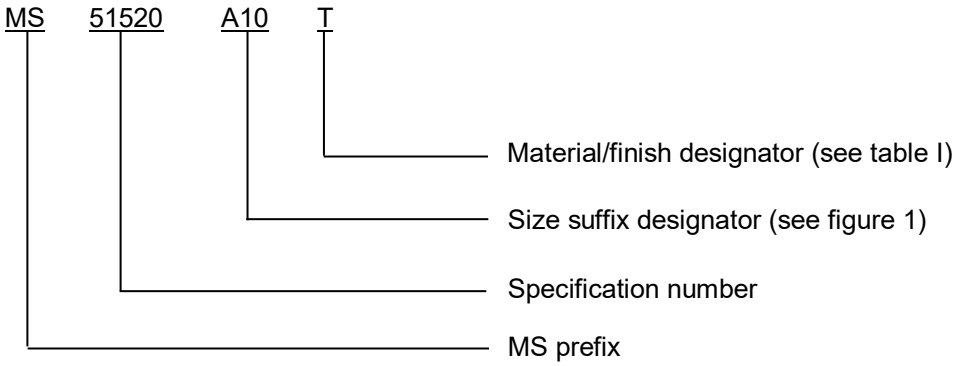
1/ Embrittlement test need not be run.
2/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
3/ Corrosion requirements shall be in accordance with MIL-PRF-32647.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook.

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5/ These finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended for Army and Marine ground systems. The use of coatings depositing hexavalent chromium for Army and Marine ground vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement. For new designs PIN codes Blank, Z, and ZC are not to be supplied to Army and Marine ground forces.
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.
7/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA. The use of cadmium plating for Army and Marine ground vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement. Pin code J or KB may be substituted for cadmium (blank PIN code) for Army ground vehicle applications.
8/ Army ground vehicle applications. MIL-PRF-32647 Type KA is intended as a hazardous free plating material and is a direct replacement for zinc plating with hexavalent chromium conversion coating, while Type KB is a hazardous free plating material and is a direct replacement for cadmium plating. Both types have equivalent torque/tension behavior as the plating they replace.
9/ Titanium shall not be used in oxygen or potable water systems.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

PIN: The PIN consists of the letters “MS”, the specification number, a letter and number for nipple size, and a letter for material finish designator.



PIN example: MS51520A10T indicates a nipple, tube, .6250 inch (15.875 mm), titanium anodized.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

For new designs PIN codes Blank, P, Z, and ZC are not to be supplied to Army and Marine ground vehicles.

Class I and II ozone depleting substances (ODS) shall not be used in MS51520 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Referenced documents shall be of the issue in effect on date of invitations for bid.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

MIL-DTL-16232
MIL-PRF-32647
MS51531
MS51533
MS51860
ASME B1.1
ASTM B117
ASTM B633
ASTM B695
ASTM F1136/F1136M
ASTM MNL37
SAE-AMS-C-81562
SAE-AMS-QQ-P-416
SAE-AMS-STD-595/36076
SAE-AMS-STD-595/36081
SAE-AMS-STD-595/36099
SAE-AMS-STD-595/36118
SAE-AMS-STD-595/36134
SAE-AMS-STD-595/36152
SAE-AMS-STD-595/36170
SAE-AMS-STD-595/36173
SAE-AMS-STD-595/36176
SAE-AMS-STD-595/36231
SAE-AMS-STD-595/36251
SAE-AMS-STD-595/36270
SAE-AMS-STD-595/36280
SAE-AMS-STD-595/36293
SAE-AMS2417
SAE-AMS2486
SAE-AMS2488
SAE-AMS2700
SAE-AS28778
SAE-J514
T9070-AL-DPC-020/077-2

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2020-054)

Review activities:
Army - AT, MI
Navy - CG, MC, SA, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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